Software
Performance
Engineering

LECTURE 17

Synchronization without Locks

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Outline

- Sequential Consistency (SC)
- Mutual Exclusion w/o Locks under SC
- Relaxed Memory Consistency
- Compare and Swap
- Lock-free Algorithms
- The ABA Problem

SEQUENTIAL CONSISTENCY

Memory Models

Initially, `a = b = 0`.

Processor 0

```
mov 1, a      ;Store
mov b, %ebx   ;Load
```

Processor 1

```
mov 1, b      ;Store
mov a, %eax   ;Load
```

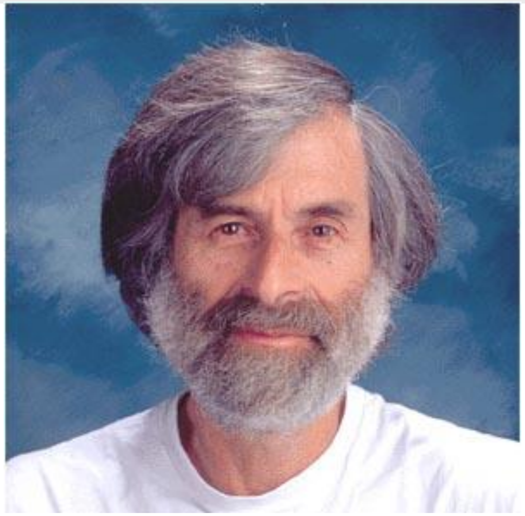
Q. Is it possible that Processor 0's `%ebx` and Processor 1's `%eax` both contain the value 0 after the processors have both executed their code?

A. It depends on the memory model: how memory operations behave in the parallel computer system.

Sequential Consistency

"The result of any execution is the same as if the operations of all the processors were executed in some sequential order, and the operations of each individual processor appear in this sequence in the order specified by its program."

— Leslie Lamport [1979]



- The sequence of instructions as defined by a processor's program are **interleaved** with the corresponding sequences defined by the other processors' programs to produce a global **linear order** of all instructions.
- A `LOAD` instruction receives the value stored to that address by the most recent `STORE` instruction that precedes the `LOAD`, according to the linear order.
- The hardware can do whatever it wants, but for the execution to be sequentially consistent, it must appear as if `LOAD`'s and `STORE`'s obey some global linear order.

Example

Initially, `a = b = 0`.

Processor 0

```
1  mov 1, a      ;Store
2  mov b, %ebx   ;Load
```

Processor 1

```
3  mov 1, b      ;Store
4  mov a, %eax   ;Load
```

	Interleavings					
	1	1	1	3	3	3
	2	3	3	1	1	4
	3	2	4	2	4	1
	4	4	2	4	2	2
`%eax`	`1`	`1`	`1`	`1`	`1`	`0`
`%ebx`	`0`	`1`	`1`	`1`	`1`	`1`

Sequential consistency implies that no execution ends with `%eax = %ebx = 0`.

Reasoning about Sequential Consistency

- An execution induces a "happens before" relation, denoted as $\rightarrow$.
- The $\rightarrow$ relation is linear (a total order), meaning that in particular for any two distinct instructions `x` and `y`, either `x` $\rightarrow$ `y` or `y` $\rightarrow$ `x`.
- The $\rightarrow$ relation respects processor order, the order of instructions in each processor.
- A `LOAD` from a location in memory reads the value written by the most recent `STORE` to that location according to $\rightarrow$.
- For the memory resulting from an execution to be sequentially consistent, there must exist such a linear order $\rightarrow$ that yields that memory state.

Mutual Exclusion w/o Locks

Mutual-Exclusion Problem

Recall

A critical section is a piece of code that accesses a shared data structure that must not be executed by two or more parallel strands (mutual exclusion).

- Computer hardware provides atomic read-modify-write instructions
  - e.g., atomic swap (X86 `xchg`), TEST-AND-SET, COMPARE-AND-SWAP, LOAD-LINKED-STORE-CONDITIONAL
- Synchronization libraries use these instructions to implement locks, but you can use them directly
  - The C library `stdatomic.h`* provides a long list of atomics for most architectures
  - LLVM & GCC provide compiler built-in functions for synchronization, but less portable

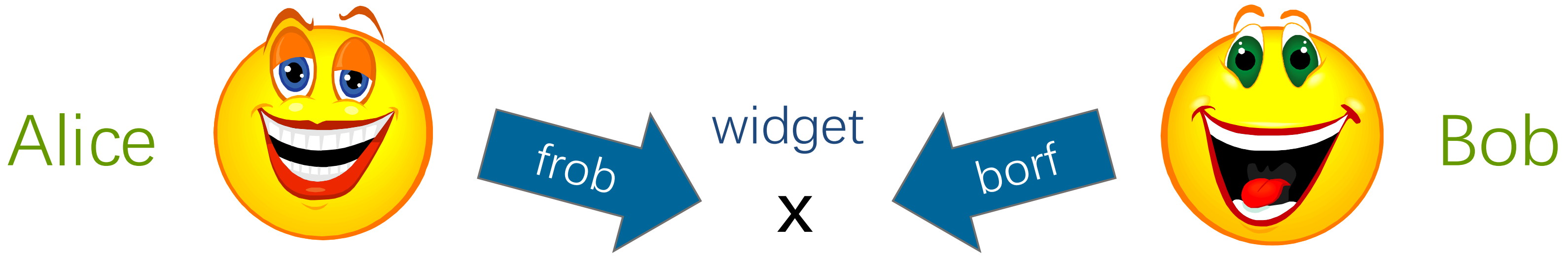
*See http://en.cppreference.com/w/c/atomic .

Mutual-Exclusion Problem

Q. Can mutual exclusion be implemented with atomic `LOAD`'s and `STORE`'s as the only memory operations?

A. Yes, Theodorus J. Dekker and Edsger Dijkstra showed that it can, at least for computers with sequentially consistent memory models.

Peterson's Algorithm




```
widget x; //protected variable
bool A_wants = false;
bool B_wants = false;
enum {A, B} turn;
```

Alice

```
A_wants = true;
turn = B;
while (B_wants && turn==B);
frob(&x); //critical section
A_wants = false;
```

Bob

```
B_wants = true;
turn = A;
while (A_wants && turn==A);
borf(&x); //critical section
B_wants = false;
```

Mutual Exclusion With Locks

- Mutual Exclusion: for any threads A and B with critical section executions
  - **`lock()`** ➔ $CS_{\{A,B\}}$ ➔ **`unlock()`**
  - either CS_A ➔ CS_B or CS_B ➔ CS_A

- Deadlock Freedom: if some thread calls **`lock()`**
  - And never returns
  - Other threads must be completing **`lock()`** and **`unlock()`** calls infinitely often

- Starvation Freedom: if some thread calls **`lock()`**
  - It will eventually return

Proof of Mutual Exclusion

Theorem. Peterson's algorithm guarantees mutual exclusion.

Proof.

- Assume by way of contradiction that both Alice and Bob execute the critical section concurrently.
- Consider the most-recent time that each of them executed the code before entering the critical section.
- Let's derive a contradiction.

Proof of Mutual Exclusion

Alice

```
A_wants = true;
turn = B;
while (B_wants && turn==B);
frob(&x); //critical section
A_wants = false;
```

Bob

```
B_wants = true;
turn = A;
while (A_wants && turn==A);
borf(&x); //critical section
B_wants = false;
```

Proof of Mutual Exclusion

Alice

```
A_wants = true;
turn = B;
while (B_wants && turn==B);
frob(&x); //critical section
A_wants = false;
```

Bob

```
B_wants = true;
turn = A;
while (A_wants && turn==A);
borf(&x); //critical section
B_wants = false;
```

- Assume WLOG that Bob was the last to write to `turn`:

$$\text{write}_A(\texttt{turn = B}) \rightarrow \text{write}_B(\texttt{turn = A})$$

Proof of Mutual Exclusion

Alice

```
A_wants = true;
turn = B;
while (B_wants && turn==B);
frob(&x); //critical section
A_wants = false;
```

Bob

```
B_wants = true;
turn = A;
while (A_wants && turn==A);
borf(&x); //critical section
B_wants = false;
```

- Assume WLOG that Bob was the last to write to `turn`:

  $\text{write}_A(\texttt{turn = B}) \rightarrow \text{write}_B(\texttt{turn = A})$

- Alice's program order:

  $\text{write}_A(\texttt{A\_wants = true}) \rightarrow \text{write}_A(\texttt{turn = B})$

Proof of Mutual Exclusion

Alice

```
A_wants = true;
turn = B;
while (B_wants && turn==B);
frob(&x); //critical section
A_wants = false;
```

Bob

```
B_wants = true;
turn = A;
while (A_wants && turn==A);
borf(&x); //critical section
B_wants = false;
```

- Assume WLOG that Bob was the last to write to `turn`:

  $\text{write}_A(\texttt{turn = B}) \rightarrow \text{write}_B(\texttt{turn = A})$

- Alice's program order:

  $\text{write}_A(\texttt{A\_wants = true}) \rightarrow \text{write}_A(\texttt{turn = B})$

- Bob's program order:

  $\text{write}_B(\texttt{turn = A}) \rightarrow \text{read}_B(\texttt{A\_wants}) \rightarrow \text{read}_B(\texttt{turn})$

Proof of Mutual Exclusion

Alice

```
A_wants = true;
turn = B;
while (B_wants && turn==B);
frob(&x); //critical section
A_wants = false;
```

Bob

```
B_wants = true;
turn = A;
while (A_wants && turn==A);
borf(&x); //critical section
B_wants = false;
```

$\text{write}_A(\texttt{turn = B}) \rightarrow \text{write}_B(\texttt{turn = A})$

$\text{write}_A(\texttt{A\_wants = true}) \rightarrow \text{write}_A(\texttt{turn = B})$

$\text{write}_B(\texttt{turn = A}) \rightarrow \text{read}_B(\texttt{A\_wants}) \rightarrow \text{read}_B(\texttt{turn})$

Proof of Mutual Exclusion

Alice

```
A_wants = true;
turn = B;
while (B_wants && turn==B);
frob(&x); //critical section
A_wants = false;
```

Bob

```
B_wants = true;
turn = A;
while (A_wants && turn==A);
borf(&x); //critical section
B_wants = false;
```

$\text{write}_A(\texttt{A\_wants = true}) \rightarrow \text{write}_A(\texttt{turn = B})$

$\text{write}_A(\texttt{turn = B}) \rightarrow \text{write}_B(\texttt{turn = A})$

$\text{write}_B(\texttt{turn = A}) \rightarrow \text{read}_B(\texttt{A\_wants}) \rightarrow \text{read}_B(\texttt{turn})$

Proof of Mutual Exclusion

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```
A_wants = true;
turn = B;
while (B_wants && turn==B);
frob(&x); //critical section
A_wants = false;
```

Bob

```
B_wants = true;
turn = A;
while (A_wants && turn==A);
borf(&x); //critical section
B_wants = false;
```

$\text{write}_A(\texttt{A\_wants = true}) \rightarrow \text{write}_A(\texttt{turn = B})$

$\text{write}_A(\texttt{turn = B}) \rightarrow \text{write}_B(\texttt{turn = A})$

$\text{write}_B(\texttt{turn = A}) \rightarrow \text{read}_B(\texttt{A\_wants}) \rightarrow \text{read}_B(\texttt{turn})$

Proof of Mutual Exclusion

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```
A_wants = true;
turn = B;
while (B_wants && turn==B);
frob(&x); //critical section
A_wants = false;
```

Bob

```
B_wants = true;
turn = A;
while (A_wants && turn==A);
borf(&x); //critical section
B_wants = false;
```

$\text{write}_A(\texttt{A\_wants = true}) \rightarrow$

$\text{write}_A(\texttt{turn = B}) \rightarrow \text{write}_B(\texttt{turn = A})$

$\text{write}_B(\texttt{turn = A}) \rightarrow \text{read}_B(\texttt{A\_wants}) \rightarrow \text{read}_B(\texttt{turn})$

Proof of Mutual Exclusion

Alice

```
A_wants = true;
turn = B;
while (B_wants && turn==B);
frob(&x); //critical section
A_wants = false;
```

Bob

```
B_wants = true;
turn = A;
while (A_wants && turn==A);
borf(&x); //critical section
B_wants = false;
```

$\text{write}_A(\texttt{A\_wants = true}) \rightarrow$

$\text{write}_A(\texttt{turn = B}) \rightarrow \text{write}_B(\texttt{turn = A})$

$\text{write}_B(\texttt{turn = A}) \rightarrow \text{read}_B(\texttt{A\_wants}) \rightarrow \text{read}_B(\texttt{turn})$

Proof of Mutual Exclusion

Alice

```
A_wants = true;
turn = B;
while (B_wants && turn==B);
frob(&x); //critical section
A_wants = false;
```

Bob

```
B_wants = true;
turn = A;
while (A_wants && turn==A);
borf(&x); //critical section
B_wants = false;
```

$\text{write}_A(\texttt{A\_wants = true}) \rightarrow$

$\text{write}_A(\texttt{turn = B}) \rightarrow$

$\text{write}_B(\texttt{turn = A}) \rightarrow \text{read}_B(\texttt{A\_wants}) \rightarrow \text{read}_B(\texttt{turn})$

Proof of Mutual Exclusion

Alice

```
A_wants = true;
turn = B;
while (B_wants && turn==B);
frob(&x); //critical section
A_wants = false;
```

Bob

```
B_wants = true;
turn = A;
while (A_wants && turn==A);
borf(&x); //critical section
B_wants = false;
```

write_A(`A_wants = true`) ➔

write_A(`turn = B`) ➔

write_B(`turn = A`) ➔ read_B(`A_wants`) ➔ read_B(`turn`)

- What did Bob read?

`A_wants: true`
`turn:    A`

} Bob should spin. Contradiction. ■

Deadlock Freedom

Alice

```
A_wants = true;
turn = B;
while (B_wants && turn==B);
frob(&x); //critical section
A_wants = false;
```

Bob

```
B_wants = true;
turn = A;
while (A_wants && turn==A);
borf(&x); //critical section
B_wants = false;
```

Theorem: Peterson's algorithm guarantees Deadlock Freedom.

- If Alice and Bob both try to enter the critical section, then whoever writes last to `turn` spins and the other progresses.
- If only Alice tries to enter the critical section, then she progresses, since `B_wants` is false.
- If only Bob tries to enter the critical section, then he progresses, since `A_wants` is false.

Starvation Freedom

Theorem: Peterson's algorithm guarantees starvation freedom.

Proof. Exercise. ■

Relaxed Memory Consistency

Memory Models Today

- No modern-day processor implements sequential consistency.
- All implement some form of relaxed consistency.
- Hardware actively reorders instructions.
- Compilers may reorder instructions too.

Instruction Reordering

```
mov 1, a        ;Store
mov b, %ebx     ;Load
```

Program Order

```
mov b, %ebx     ;Load
mov 1, a        ;Store
```

Execution Order

Q. Why might the hardware or compiler decide to reorder these instructions?

A. To obtain higher performance by covering load latency — instruction-level parallelism.

Instruction Reordering

```
mov 1, a        ;Store
mov b, %ebx     ;Load
```

Program Order

```
mov b, %ebx     ;Load
mov 1, a        ;Store
```

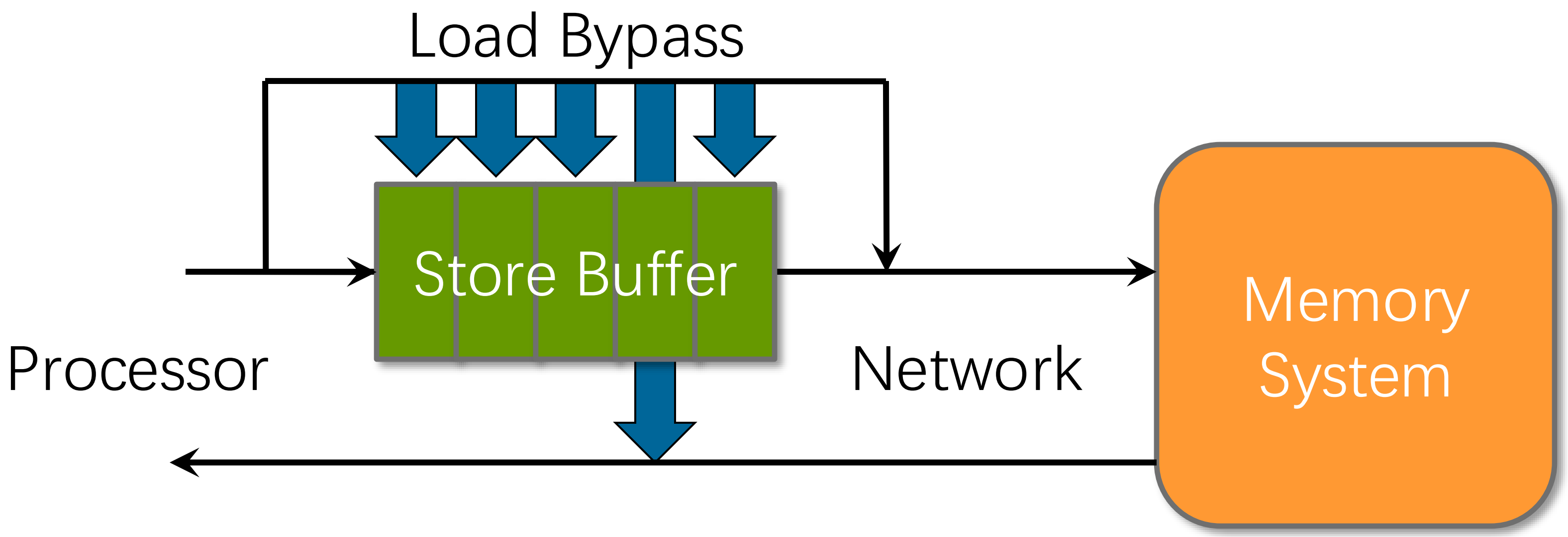
Execution Order

Q. When is it safe for the hardware or compiler to perform this reordering?

A. When a $\neq$ b.

A′. And there’s no concurrency.

Hardware Reordering



- The processor can issue **STORE**’s faster than the network can handle them ⇒ store buffer.
- Since a **LOAD** can stall the processor until it is satisfied, loads take priority, bypassing the store buffer.
- If a **LOAD** address matches an address in the store buffer, the store buffer returns the result.
- Thus, a **LOAD** can bypass a **STORE** to a different address.

x86-64 Total Store Order

Instruction Trace

~~Store1~~

~~Store2~~

Load1

~~Load2~~

~~Store3~~

Store4

Load3

~~Load4~~

~~Load5~~

Locally:

1. LOAD's are not reordered with LOAD's.
2. STORE's are not reordered with STORE's.
3. STORE's are not reordered with prior LOAD's.
4. A LOAD may be reordered with a prior STORE to a different location but not with a prior STORE to the same location.

x86-64 Total Store Order

Instruction Trace

Store1
Store2
Load1
Load2
Store3
Store4
Load3
Load4
Load5

Locally:

1. LOAD's are not reordered with LOAD's.
2. STORE's are not reordered with STORE's.
3. STORE's are not reordered with prior LOAD's.
4. A LOAD may be reordered with a prior STORE to a different location but not with a prior STORE to the same location.
5. LOAD's and STORE's are not reordered with LOCK instructions.

Globally:

6. STORE's to the same location respect a global total order.
7. LOCK instructions respect a global total order.
8. Memory ordering preserves transitive visibility ("causality").

x86-64 Total Store Order

Instruction Trace

Store1
Store2
Load1
Load2
Store3
Store4
Load3
Load4
Load5

Locally:

1. LOAD's are not reordered with LOAD's.
2. STORE's are not reordered with STORE's.
3. STORE's are not reordered with prior LOAD's.
4. A STORE to a different lo the same location.
5. L ith LOCK instructions.

Total Store Ordering (TSO) is weaker than sequential consistency.

Globally:

6. STORE's to the same location respect a global total order.
7. LOCK instructions respect a global total order.
8. Memory ordering preserves transitive visibility ("causality").

Impact of Reordering

Processor 0

```
1 mov 1, a        ;Store
2 mov b, %ebx     ;Load
```

Processor 1

```
3 mov 1, b        ;Store
4 mov a, %eax     ;Load
```

Impact of Reordering

Processor 0

```
1 mov 1, a      ;Store
2 mov b, %ebx   ;Load
```

```
2 mov b, %ebx   ;Load
1 mov 1, a      ;Store
```

Processor 1

```
3 mov 1, b      ;Store
4 mov a, %eax   ;Load
```

```
4 mov a, %eax   ;Load
3 mov 1, b      ;Store
```

The ordering ⟨2, 4, 1, 3⟩ produces `%eax = %ebx = 0`.

Instruction reordering violates sequential consistency!

Further Impact of Reordering

Peterson's algorithm revisited

Alice

```
A_wants = true;
turn = B;
while (B_wants && turn==B);
frob(&x); //critical section
A_wants = false;
```

Bob

```
B_wants = true;
turn = A;
while (A_wants && turn==A);
borf(&x); //critical section
B_wants = false;
```

- The `LOAD`'s of `B_wants` and `A_wants` can be reordered before the `STORE`'s of `A_wants` and `B_wants`, respectively
- Both Alice and Bob might enter their critical sections simultaneously!

Memory Fences

- A memory fence (or memory barrier) is a hardware action that enforces an ordering constraint between the instructions before and after the fence

- A memory fence can be issued explicitly as an instruction (x86: `mfence`) or be performed implicitly by locking, exchanging, and other synchronizing instructions

- The OpenCilk/LLVM compiler implements a memory fence via the function `atomic_strand_fence()` defined in the C header file `stdatomic.h`.*

- The typical cost of a memory fence is comparable to that of an L2-cache access

*See http://en.cppreference.com/w/c/atomic .

Restoring Consistency

Alice

```
A_wants = true;
turn = B;
while (B_wants && turn==B);
frob(&x); //critical section
A_wants = false;
```

Bob

```
B_wants = true;
turn = A;
while (A_wants && turn==A);
borf(&x); //critical section
B_wants = false;
```

Memory fences can restore sequential consistency.

```
A_wants = true;
turn = B;
atomic_thread_fence();
while (B_wants && turn==B);
frob(&x); //critical section
A_wants = false;
```

```
B_wants = true;
turn = A;
atomic_thread_fence();
while (A_wants && turn==A);
borf(&x); //critical section
B_wants = false;
```

Well, sort of.

You also need to make sure that the compiler doesn’t screw you over.

Restoring Consistency

```
widget x; //protected variable
_Atomic bool A_wants = false;
_Atomic bool B_wants = false;
_Atomic enum {A, B} turn;
```

Alice

```
atomic_store(&A_wants, true);
atomic_store(&turn, B);
while (atomic_load(&B_wants) &&
       atomic_load(&turn)==B);
frob(&x); //critical section
atomic_store(&A_wants, false);
```

Bob

```
atomic_store(&B_wants, true);
atomic_store(&turn, A);
while (atomic_load(&A_wants) &&
       atomic_load(&turn)==A);
borf(&x); //critical section
atomic_store(&B_wants, false);
```

In addition to the memory fence

- you must declare variables as `volatile` to prevent the compiler from optimizing away memory references;
- you need compiler fences around `frob()` and `borf()` to prevent compiler reordering

Restoring Consistency with C11

```
widget x; //protected variable
_Atomic bool A_wants = false;
_Atomic bool B_wants = false;
_Atomic enum {A, B} turn;
```

Alice

```
atomic_store(&A_wants, true);
atomic_store(&turn, B);
while (atomic_load(&B_wants) &&
       atomic_load(&turn)==B);
frob(&x); //critical section
atomic_store(&A_wants, false);
```

Bob

```
atomic_store(&B_wants, true);
atomic_store(&turn, A);
while (atomic_load(&A_wants) &&
       atomic_load(&turn)==A);
borf(&x); //critical section
atomic_store(&B_wants, false);
```

The C11 language standard defines its own weak memory model, in which you can control hardware and compiler reordering of memory operations by

- declaring variables as `_Atomic`; and
- using the functions `atomic_load()`, `atomic_store()`, etc. as needed

See http://en.cppreference.com/w/c/atomic

Implementing General Mutexes

Theorem [Burns-Lynch]. Any n-thread deadlock-free mutual-exclusion algorithm using only `LOAD` and `STORE` memory operations requires $\Omega(n)$ space

Theorem [Attiya et al.]: Any n-thread deadlock-free mutual-exclusion algorithm on a modern machine must use an expensive operation such as a memory fence or an atomic COMPARE-AND-SWAP operation.

Thus, hardware designers are justified when they implement special operations to support atomicity.

Compare-and-Swap

The Lock-Free Toolbox

Memory operations

- LOAD
- STORE
- CAS (COMPARE-AND-SWAP)

Compare-and-Swap

Specification

```
bool CAS(T *x, T old, T new) {
  if (*x == old) {
    *x = new;
    return true;
  }
  return false;
}
```

- Executes atomically.
- Implicit fence.

- COMPARE-AND-SWAP is provided by the **cmpxchg** instruction on x86-64
- The C header file **stdatomic.h** provides **CAS** via the built-in function

  atomic_compare_exchange_strong()

  which can operate on various integer types.*

* See http://en.cppreference.com/w/cpp/atomic/atomic_compare_exchange .

Mutex Using CAS

Theorem. An n-thread deadlock-free mutual-exclusion algorithm using CAS can be implemented using $\Theta(1)$ space.

Is it free?

Proof.

```
void lock(int *lock_var) {
  while (!CAS(*lock_var, false, true));
}
```

Yes? Acquire it

```
void unlock(int *lock_var) {
      *lock_var = false;
}
```

release it

Just the space for the mutex itself. ■

Mutex Using CAS

Theorem. An `n`-thread deadlock-free mutual-exclusion algorithm using `CAS` can be implemented using $\Theta(1)$ space.

Theorem [Burns-Lynch]. Any `n`-thread deadlock-free mutual-exclusion algorithm using only `LOAD` and `STORE` memory operations requires $\Omega(n)$ space

Summing Problem

```
int compute(const X& v);
int main() {
  const int n = 1000000;
  extern X myArray[n];
  // ...

  int result = 0;
  cilk_for (int i = 0; i < n; ++i) {
    result += compute(myArray[i]);
  }
  printf( "The result is: %f\n", result );
  return 0;
}
```

Race!

Mutex Solution

```
int compute(const X& v);
int main() {
  const int n = 1000000;
  extern X myArray[n];
  mutex L;
  // ...

  int result = 0;
  cilk_for (int i = 0; i < n; ++i) {
    int temp = compute(myArray[i]);
    L.lock();
    result += temp;
    L.unlock();
  }
  printf( "The result is: %f\n”, result );
  return 0;
}
```

Mutex Solution

Yet all we want is to atomically execute a LOAD of x followed by a store of x.

```
int compute(const X& v);
int main() {
  const int n = 1000000;
  extern X myArray[n];
  mutex L;
  // ...

  int result = 0;
  cilk_for (int i = 0; i < n; ++i) {
    int temp = compute(myArray[i]);
    L.lock();
    result += temp;
    L.unlock();
  }
  printf( "The result is: %f\n”, result );
  return 0;
}
```

Q. What happens if the OS swaps out a loop iteration just after it acquires the mutex?

A. All other loop iterations must wait.

CAS Solution

```
int result = 0;
cilk_for (int i = 0; i < n; ++i) {
   int temp = compute(myArray[i]);
   int old, new;
   do {
      old = result;
      new = old + temp;
   } while (!CAS(&result, old, new));
}
```

Q. Now what happens if the OS swaps out a loop iteration?

A. No other loop iteration needs to wait. The algorithm is **nonblocking**

LOCK-FREE ALGORITHMS

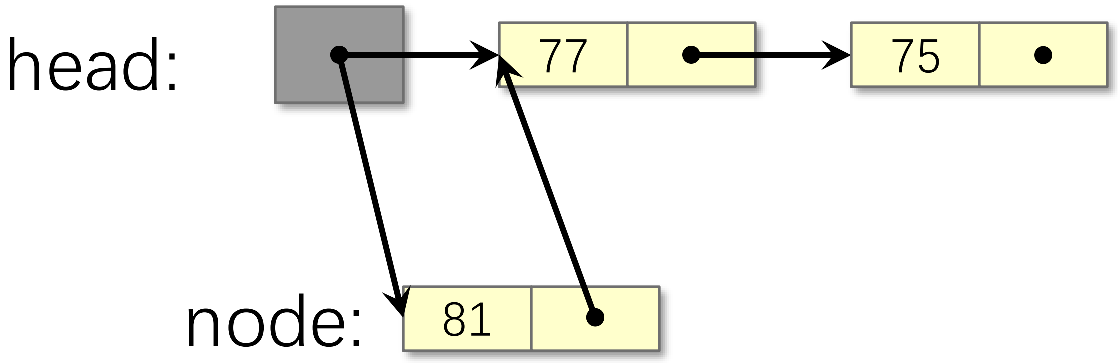
Lock-Free Stack

```
struct Node {
  Node* next;
  int data;
};

struct Stack {
    Node* head;
        ⋮
```

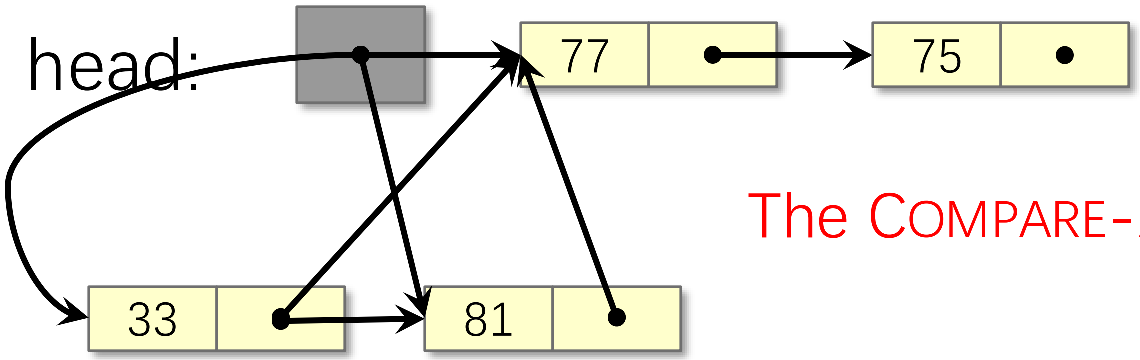
Lock-Free PUSH

```
void push(Node* node) {
  do {
        node->next = head;
     } while (!CAS(&head, node->next, node));
}
```



Lock-Free PUSH with Contention

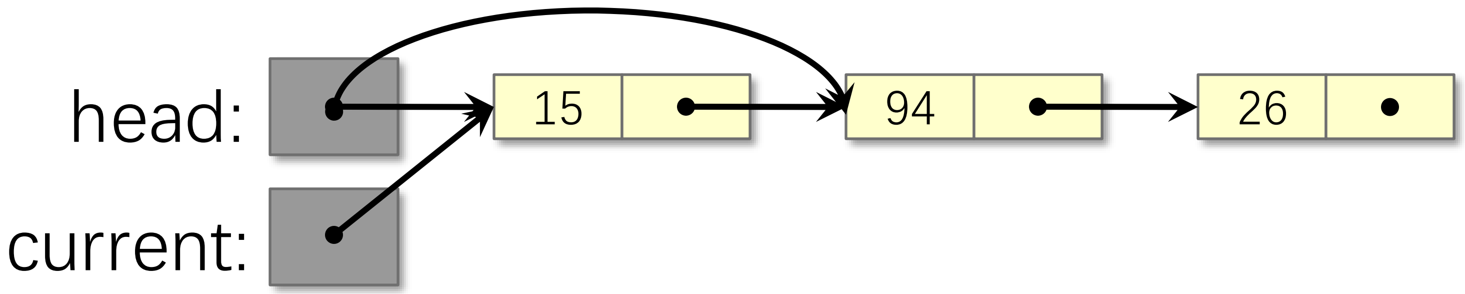
```
void push(Node* node) {
  do {
        node->next = head;
     } while (!CAS(&head, node->next, node));
}
```



The COMPARE-AND-SWAP fails!

Lock-Free POP

```
Node* pop() {
  Node* current = head;
  while (current) {
    if (CAS(&head, current, current->next)) break;
    current = head;
  }
  return current;
}
```



Performance Considerations

COMPARE-AND-SWAP acquires a cache line in exclusive mode, invalidating the cache line in other caches.

- **Result**: High contention if all processors CAS to same cache line.

Better

- First, read the memory location to check whether the value changed before attempting a CAS.
- Only CAS if the value didn't change.

Similar to the test and spin lock that reads the lock status before attempting the `xchg` operation.

Lock-Free Push and Pop

```
void push(Node* node) {
  do {
        node->next = head;
      } while (head != node->next ||
               !CAS(&head, node->next, node));
}
```

```
Node* pop() {
  Node* current = head;
  while (current) {
    if (head == current &&
        CAS(&head, current, current->next)) break;
    current = head;
  }
  return current;
}
```

Even better: add exponential backoff

Lock-Free Data Structures

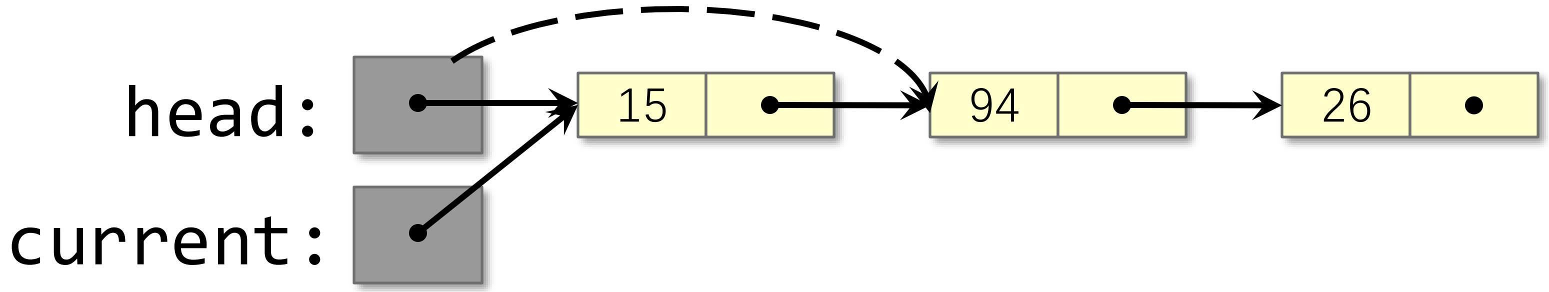
- Efficient lock-free algorithms are known for a variety of classical data structures (e.g., linked lists, queues, skip lists, hash tables)
- In theory, a thread might starve. Because of contention, its operation might never complete. In practice, starvation rarely happens
- Transactional memory offers another way to mutual exclusion w/o locks
  - TM allows a block of code to execute atomically without worrying about locks or complicated lock-free protocols

Practical issues with lock-free programming

- memory management
- contention
- the ABA problem

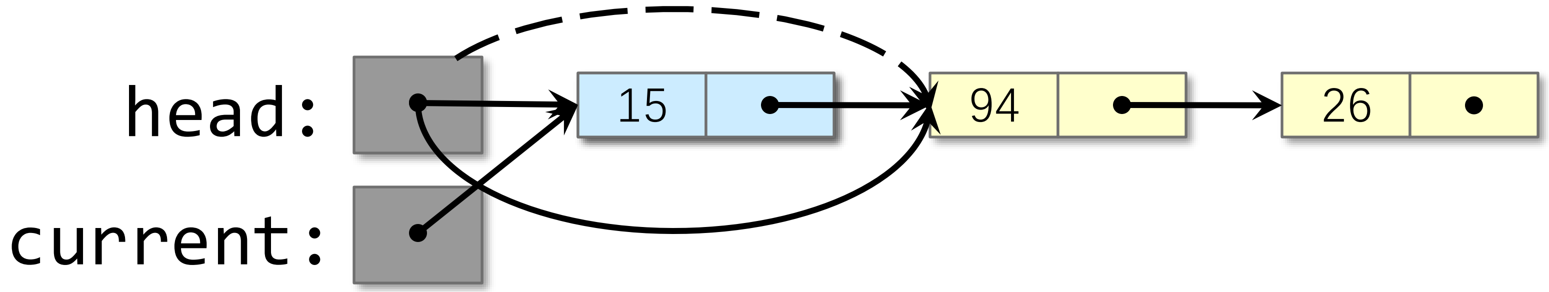
The ABA Problem

ABA Example



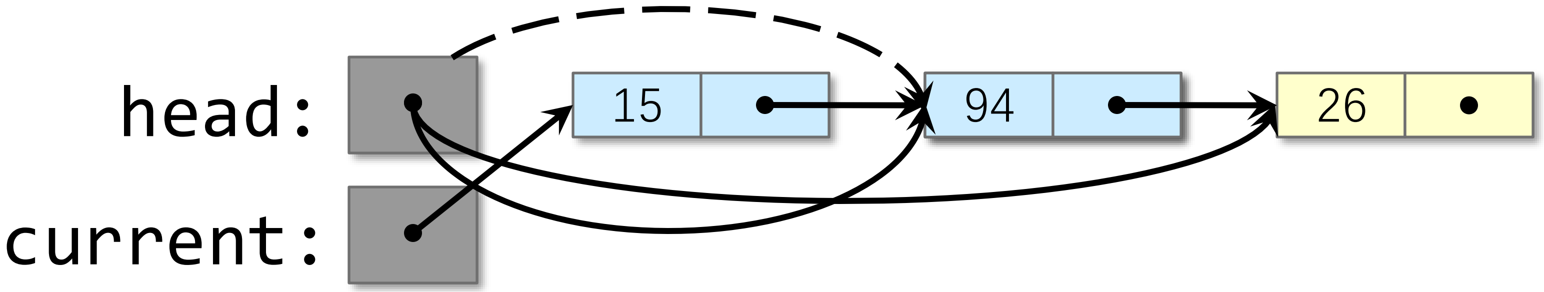
1. Strand 1 begins to pop the node containing `15`, but stalls after reading `current->next`.

ABA Example



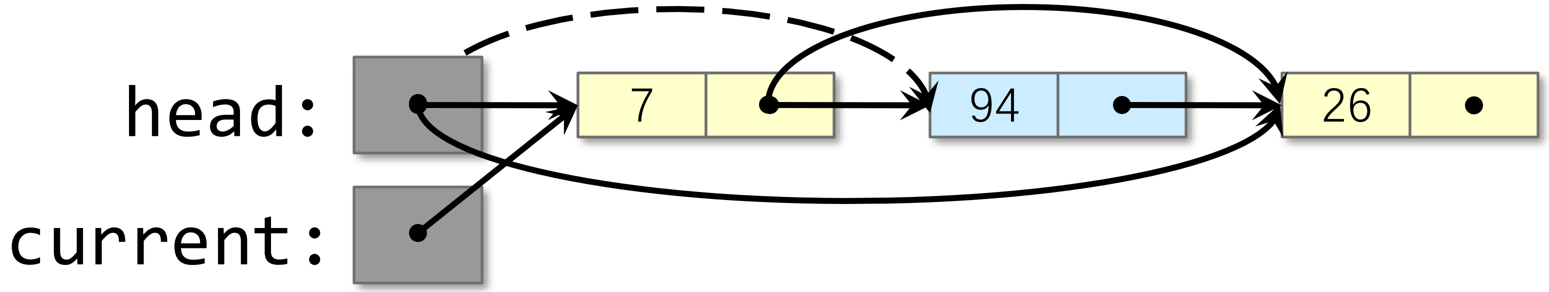
1. Strand 1 begins to pop the node containing `15`, but stalls after reading `current->next`.
2. Strand 2 pops the node containing `15`.

ABA Example



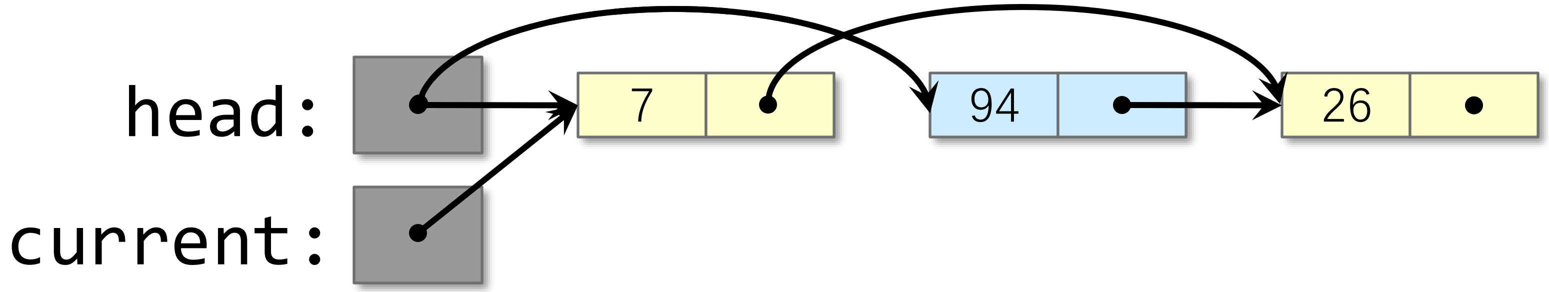
1. Strand 1 begins to pop the node containing `15`, but stalls after reading `current->next`.
2. Strand 2 pops the node containing `15`.
3. Strand 2 pops the node containing `94`.

ABA Example



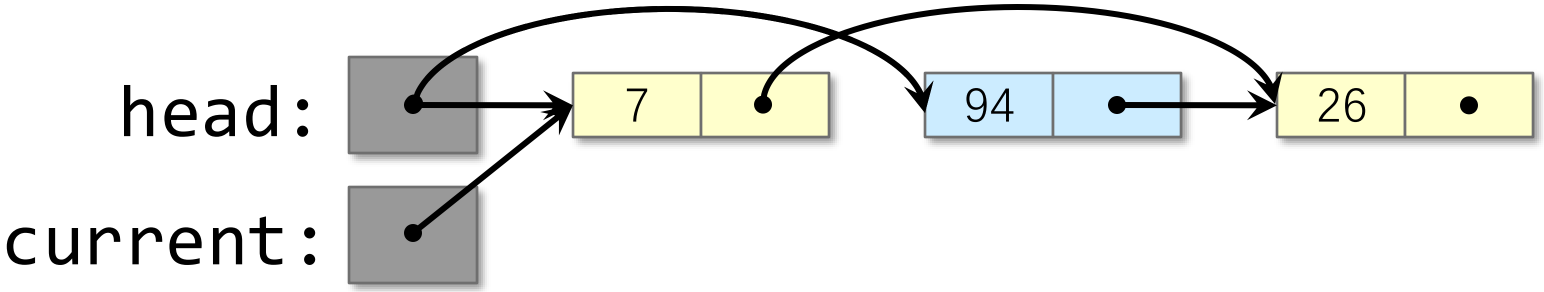
1. Strand 1 begins to pop the node containing 15, but stalls after reading `current->next`.
2. Strand 2 pops the node containing 15.
3. Strand 2 pops the node containing 94.
4. Strand 2 pushes the value 7, reusing the node that contained 15.

ABA Example



1. Strand 1 begins to pop the node containing `15`, but stalls after reading `current->next`.
2. Strand 2 pops the node containing `15`.
3. Strand 2 pops the node containing `94`.
4. Strand 2 pushes the node `7`, reusing the node that contained `15`.
5. Strand 1 resumes, and its `CAS` succeeds, removing `7`, but putting garbage back on the list.

ABA Example



So we went from the head being

1. A = node with 15
2. To B = node with 94
3. Back to A = node that had 15 (but now has 7)

Looks the same but the world changed completely in the interim…

Notice that checking the value in the node will not help since in step 3 we could have inserted 15 back and gotten the same error…

Solutions to ABA

Versioning

- Pack a version # with each pointer in the same atomically updatable word
- Increment the version number every time the pointer is changed
- CAS both the pointer and the version # as a single atomic operation

Issue

- Version numbers may need to be very large.

Reclamation

- Prevent node reuse while pending requests exist.
- e.g. prevent node 15 from being reused as 7 while Strand 1 still executing

The ABBA lock-free Approach